

Exercise Sheet 12e

Predicate Logic – Natural Deduction & Semantics

Consider the following signature:

- Function symbols: **zero** (arity 0); **succ** (arity 1)
- Predicate symbols: $<$ (arity 2); $\leq$ (arity 2)

We will use infix notation for the binary symbols $<$ and $\leq$. For simplicity we write 0 for **zero**, 1 for **succ(zero)**, 2 for **succ(succ(zero))**, etc. Consider the following formulas that capture properties of the above symbols:

- let S_1 be $\neg\exists x.x < 0$
- let S_2 be $\forall x.\exists y.x < y \vee y < x$

1. Provide a constructive Natural Deduction proof of $(S_1) \rightarrow \forall x.\neg x < 0$
2. Provide a Constructive Natural Deduction proof of $(S_1) \rightarrow (S_2) \rightarrow \exists x.0 < x$
3. Provide a model M_1 such that $\models_{M_1} S_2$.
4. Provide a model M_2 such that $\models_{M_2} \neg S_2$.